PAINTINGS KNOWLEDGE GRAPH

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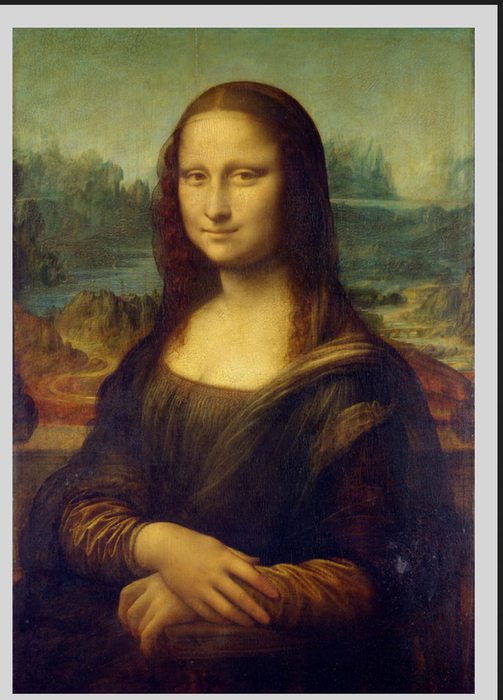

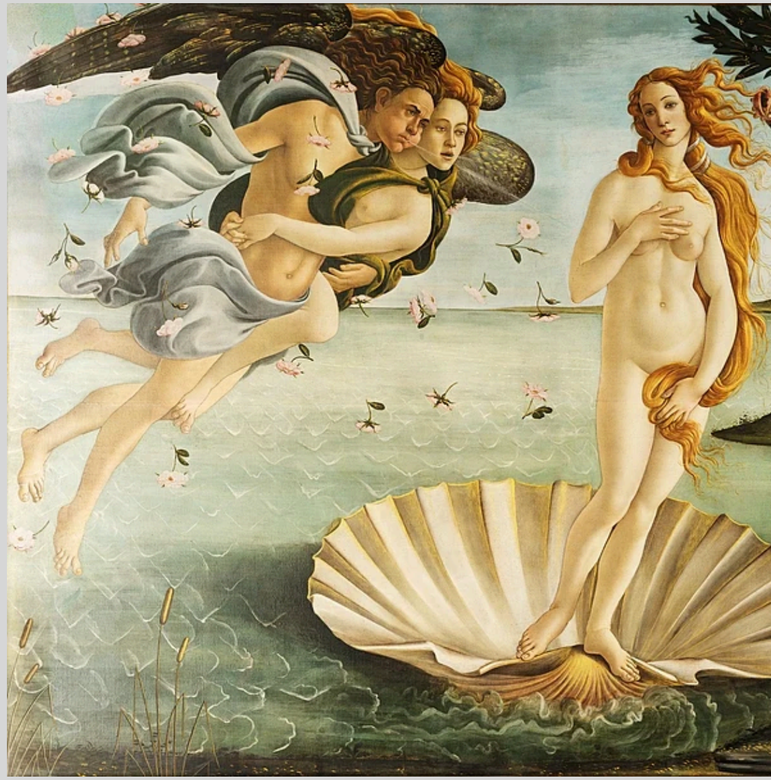

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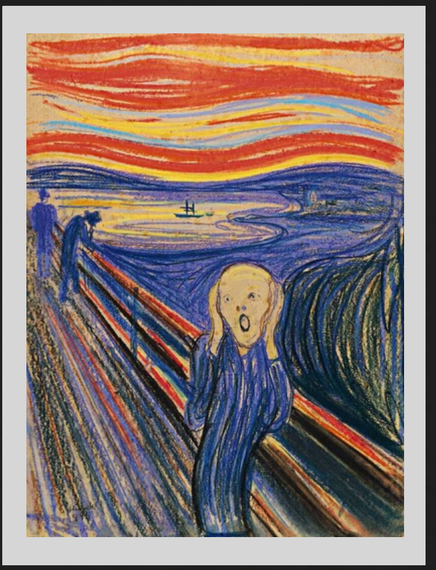


Objectives

This project serves as a valuable resource for building a graph database that contains detailed information about paintings, whether for research purposes or general knowledge. Through data collection, KG modelling, and hosting, we have built a knowledge graph with the data related to paintings, answered research questions, and visualized query results.

Tools

Source: Wikidata
Mapping: Python (SPARQLWrapper/RDFLib)
Modeling: Protégé
KG hosting: GraphDB
Query: SPARQL

Data Collection

The data was collected using Wikidata SPARQL Endpoint. In order to have a more detailed structure and to obtain more information, we relied on 3 CONSTRUCT queries to obtain triples associated with each class. The classes are as follow: Painting, Artist, and Museum.
For each class we obtain the following results:

- Paintings: 62961 rows
- Artists: 12971 rows
- Museums: 11929 rows

Where the triples not only refer to the names of the retrieved entity, but also to the relation between the entity and its properties such as date, location, etc.

Painting:

```
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

CONSTRUCT {
  ?painting rdfs:label ?paintingLabel .
  ?painting <http://example.org/artist> ?artist .
  ?painting <http://example.org/artistName> ?artistLabel .
  ?painting <http://example.org/year> ?year .
  ?painting <http://example.org/museum> ?museum .
  ?painting <http://example.org/museumName> ?museumLabel .
  ?painting <http://example.org/movementName> ?movementLabel .
  ?painting <http://example.org/materialName> ?materialLabel .
  ?painting <http://example.org/imageURL> ?image
}
WHERE {
  ?painting wdt:P31 wd:Q3305213 .
  ?painting wdt:P170 ?artist .
  ?painting wdt:P571 ?year .
  ?painting wdt:P195 ?museum .
  ?painting wdt:P135 ?movement .
  ?painting wdt:P186 ?material .
  ?painting wdt:P18  ?image .

  ?painting rdfs:label ?paintingLabel .
  FILTER(LANG(?paintingLabel) = "en")

  ?artist rdfs:label ?artistLabel .
  FILTER(LANG(?artistLabel) = "en")

  ?museum rdfs:label ?museumLabel .
  FILTER(LANG(?museumLabel) = "en")

  ?movement rdfs:label ?movementLabel .
  FILTER(LANG(?movementLabel) = "en")

  ?material rdfs:label ?materialLabel .
  FILTER(LANG(?materialLabel) = "en")
}
LIMIT 50000
```

Artist:

```
CONSTRUCT {
  ?artist rdfs:label ?artistLabel.
  ?artist schema:birthDate ?birthDate.
  ?artist schema:birthPlace ?birthPlaceLabel.
  ?artist schema:birthCountry ?birthCountryLabel.
} WHERE {
  ?painting wdt:P31 wd:Q3305213.        # Painting
  ?painting wdt:P170 ?artist.           # Created by artist

  ?artist rdfs:label ?artistLabel.
  FILTER(LANG(?artistLabel) = "en")     # English artist label

  ?artist wdt:P569 ?birthDate.          # Birth date
  ?artist wdt:P19 ?birthPlace.          # Birthplace
  ?birthPlace rdfs:label ?birthPlaceLabel.
  FILTER(LANG(?birthPlaceLabel) = "en") # English place label

  ?birthPlace wdt:P17 ?birthCountry.    # Country of place
  ?birthCountry rdfs:label ?birthCountryLabel.
  FILTER(LANG(?birthCountryLabel) = "en") # English country label
}
LIMIT 3500
```

Museum:

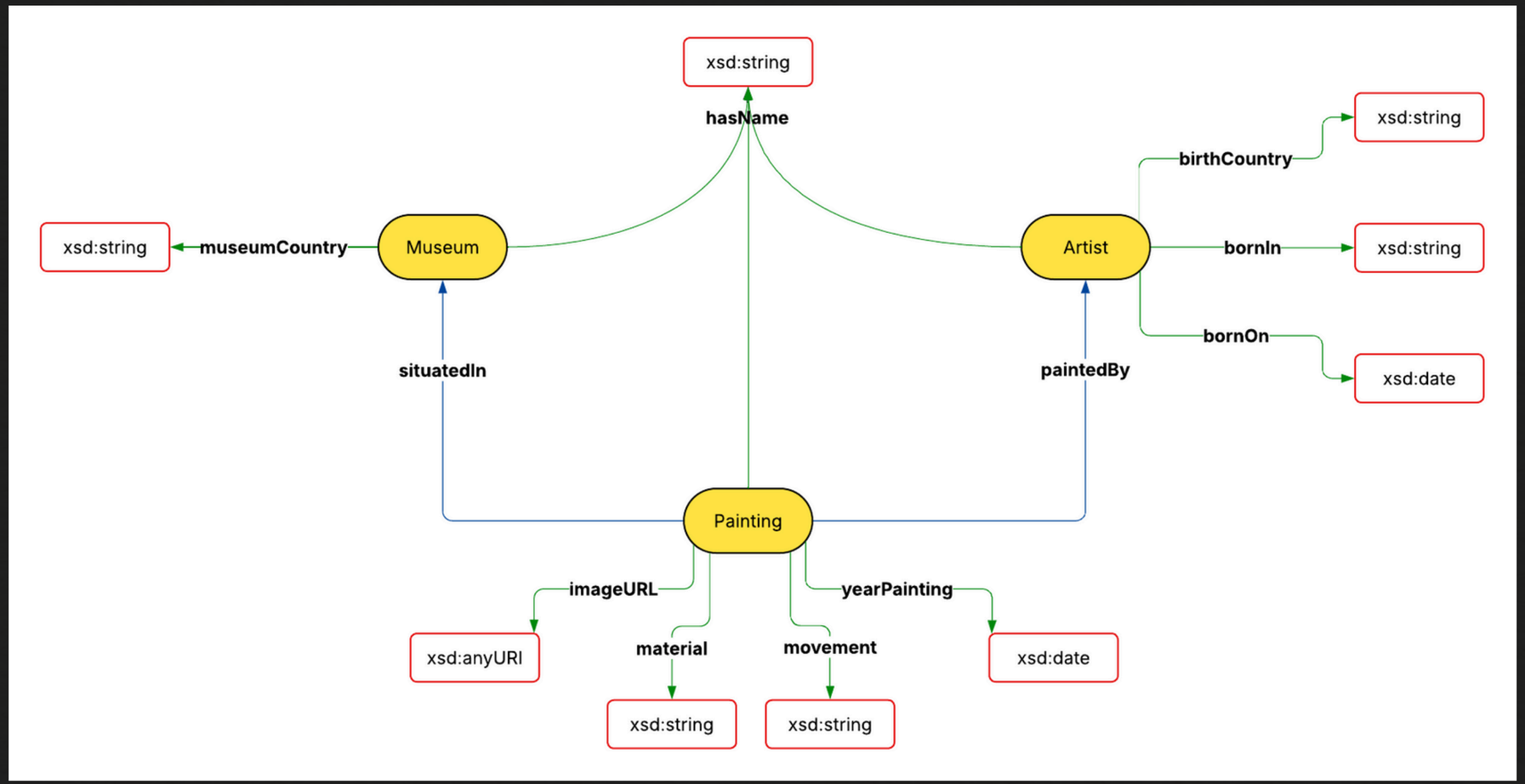
```
CONSTRUCT {
  ?museum rdfs:label ?museumLabel .
  ?museum <http://example.org/country> ?museumCountryLabel
}
WHERE {
  ?museum wdt:P31 wd:Q207694 . # museum
  ?museum rdfs:label ?museumLabel .
  FILTER(LANG(?museumLabel) = "en")

  ?museum wdt:P17 ?museumCountry .
  ?museumCountry rdfs:label ?museumCountryLabel .
  FILTER(LANG(?museumCountryLabel) = "en")
}
```

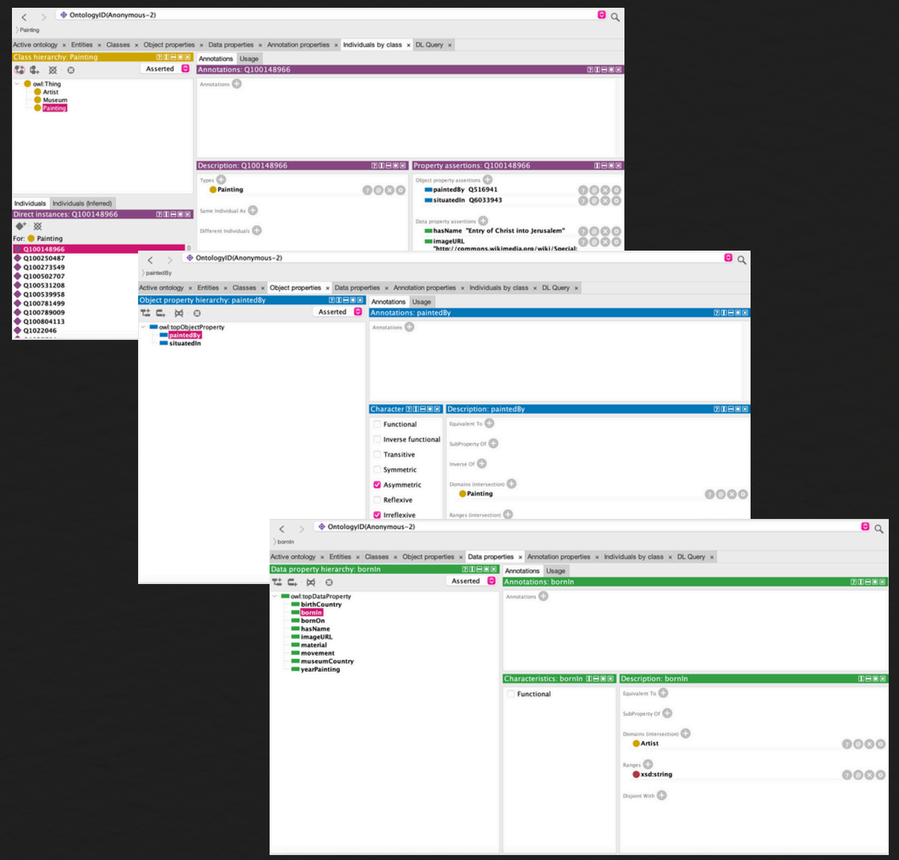
Mapping of the dataset

Due to a limitation of wikidata, the data has been retrieved leveraging a Colab notebook.
The SPARQLwrapper library was set up to retrive the query results by connecting to the wikidata endpoint and the data was then converted into an RDF/XML format. The results were then loaded and transformed into a graph by relying on the RDFLib library.

In order to create the specific structure required of the data and to populate the graph in RDFLib we were first required to model the knowledge graph in Protégé.

Modeling the KG–Ontology

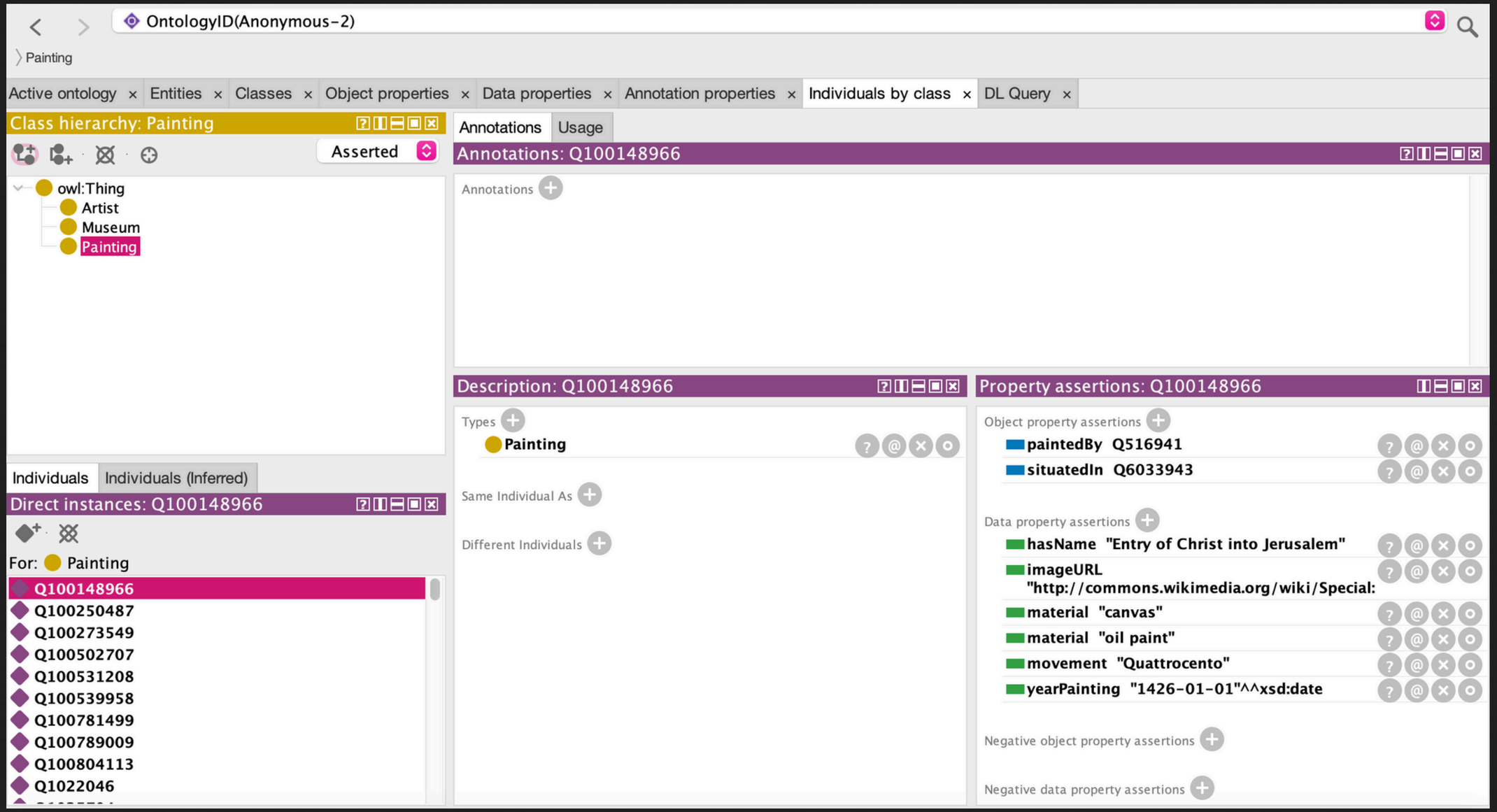


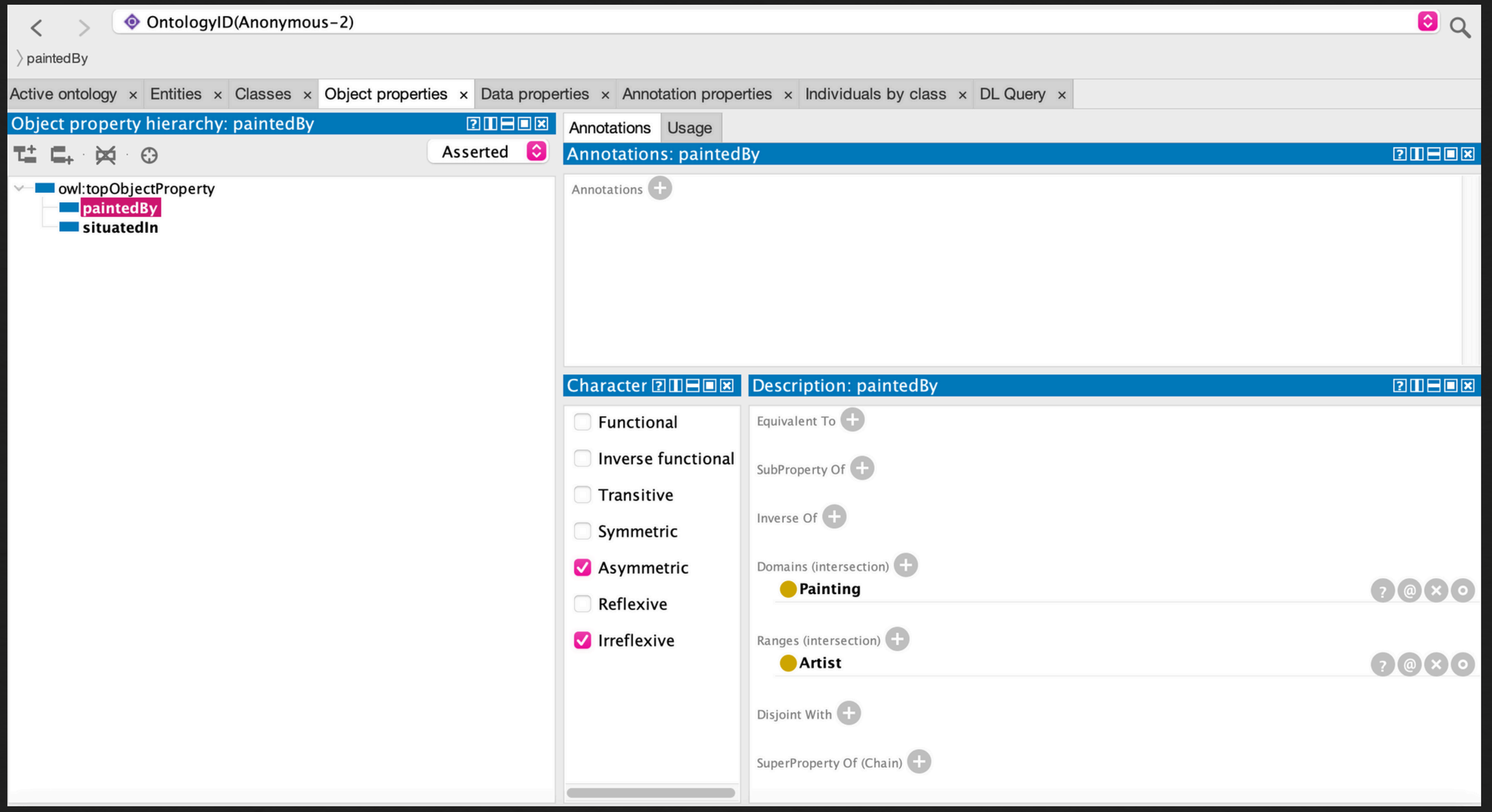


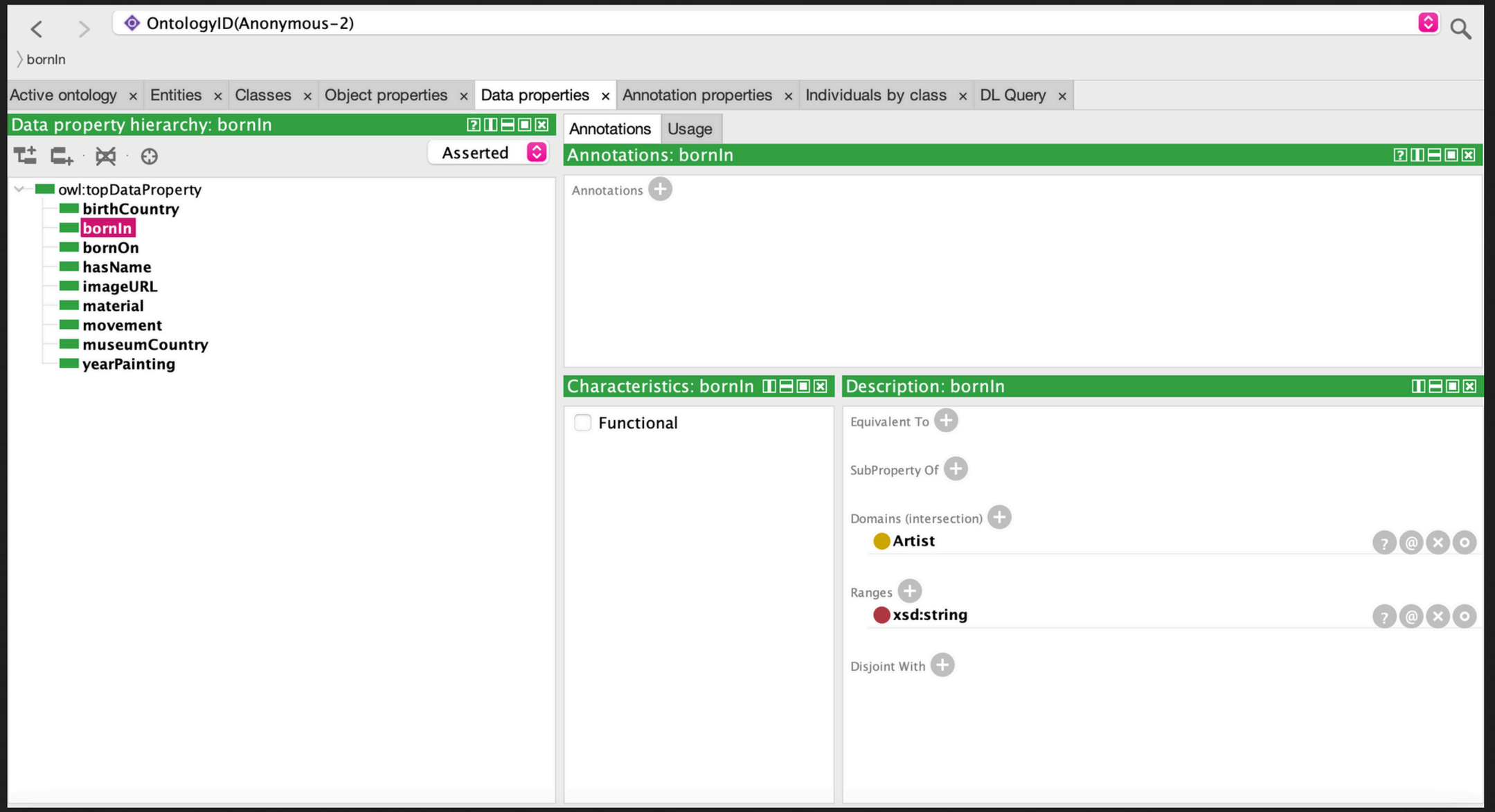
Modeling the KG

We relied on Protégé to allow us to set down the structure of the Knowlege graph. In Protégé, we defined the classes, data properties and object properties. Additionally, the ranges, the domains and the characteristics. This allowed us to have the full picture of how the knowledge graph should be structured.

Following the structure we defined in Protégé, we relied on the RDFLib library to build the final graph and populate it with the data we have collected.

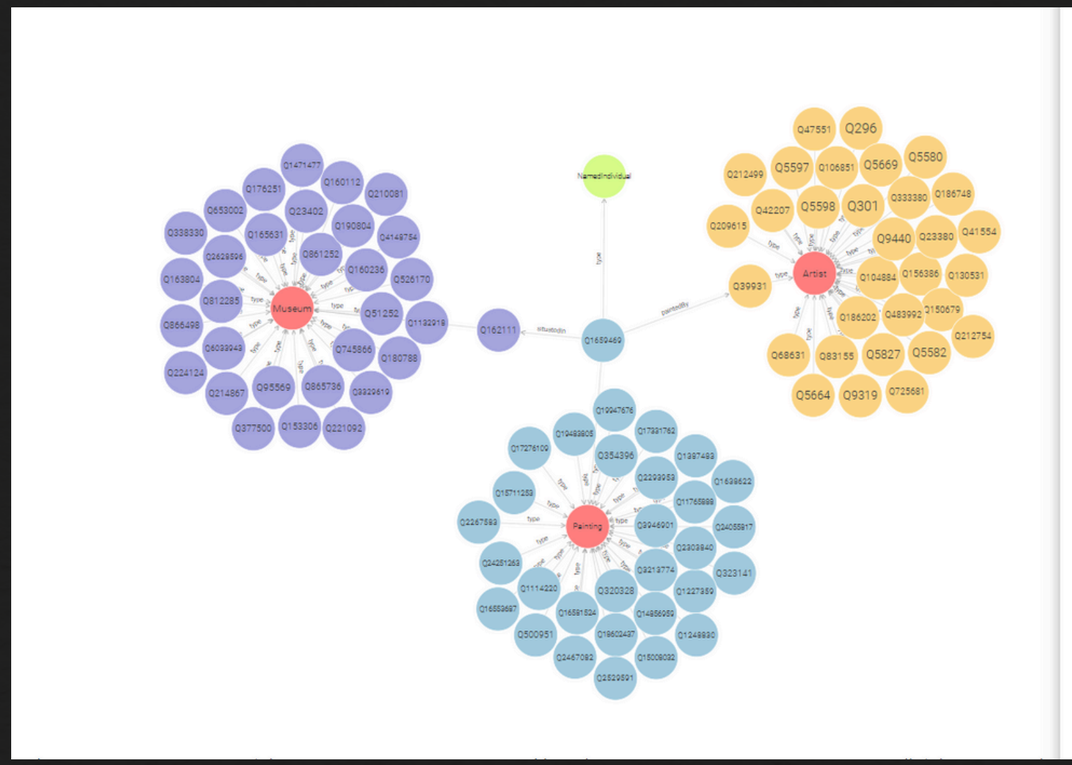

OntologyID(Anonymous-2)
Painting
Active ontology
Entities
Classes
Object properties
Data properties
Annotation properties
Individuals by class
DL Query
Class hierarchy: Painting
Asserted
owl:Thing
Artist
Museum
Painting
Annotations
Usage
Annotations: Q100148966
Annotations
Individuals
Individuals (Inferred)
Direct instances: Q100148966
For: Painting
Q100148966
Q100250487
Q100273549
Q100502707
Q100531208
Q100539958
Q100781499
Q100789009
Q100804113
Q1022046
Description: Q100148966
Types
Painting
Same Individual As
Different Individuals
Property assertions: Q100148966
Object property assertions
paintedBy Q516941
situatedIn Q6033943
Data property assertions
hasName "Entry of Christ into Jerusalem"
imageURL "http://commons.wikimedia.org/wiki/Special:
material "canvas"
material "oil paint"
movement "Quattrocento"
yearPainting "1426-01-01"^^xsd:date
Negative object property assertions
Negative data property assertions


OntologyID(Anonymous-2)
paintedBy
Active ontology
Entities
Classes
Object properties
Data properties
Annotation properties
Individuals by class
DL Query
Object property hierarchy: paintedBy
Asserted
owl:topObjectProperty
paintedBy
situatedIn
Annotations
Usage
Annotations: paintedBy
Annotations
Character
Functional
Inverse functional
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Description: paintedBy
Equivalent To
SubProperty Of
Inverse Of
Domains (intersection)
Painting
Ranges (intersection)
Artist
Disjoint With
SuperProperty Of (Chain)


OntologyID(Anonymous-2)
bornIn
Active ontology
Entities
Classes
Object properties
Data properties
Annotation properties
Individuals by class
DL Query
Data property hierarchy: bornIn
Asserted
owl:topDataProperty
birthCountry
bornIn
bornOn
hasName
imageURL
material
movement
museumCountry
yearPainting
Annotations
Usage
Annotations: bornIn
Annotations
Characteristics: bornIn
Functional
Description: bornIn
Equivalent To
SubProperty Of
Domains (intersection)
Artist
Ranges
xsd:string
Disjoint With

Hosting using GraphDB

In order to query and visualize the graph, we relied on GraphDB. We can see the graph structure:



Q1659469

Q1659469

Types:

ex:Painting

owl:NamedIndividual

RDF Rank:

0

Search instance properties

ex:hasName
In the Summer

ex:imageURL
http://commons.wikimedia.org/wiki/Special:FilePath/Auguste%20Renoir%20-%20En%20%C3%A9t%C3%A9%20-%20La%20boh%C3%A9mienne%20-%20Google%20Art%20Project.jpg xsd:anyURI

ex:material
oil paint Show 1 more

ex:movement
Impressionism Show 3 more

Q39931

Q39931

Types:

ex:Artist

owl:NamedIndividual

RDF Rank:

0

ex:birthCountry
France

ex:bornIn
Limoges

ex:bornOn
1841-02-25 xsd:date

ex:hasName
Pierre-Auguste Renoir

Q162111

Q162111

Types:

ex:Museum

owl:NamedIndividual

RDF Rank:

0

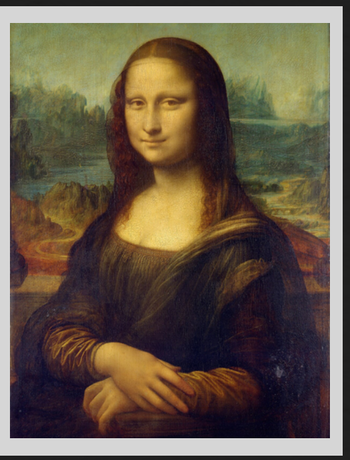
ex:hasName
Alte Nationalgalerie

ex:museumCountry
Germany

Query 1

Which foreign museums contains Italian paintings?

Mona Lisa



Leonardo Da Vinci
Louvre Museum
France

Entry of Christ into Jerusalem

Alessandro Magnasco
Museum of Fine Arts of Córdoba
Spain

Allegory of Music (Erato)

Filippino Lippi
Gemäldegalerie
Germany

```
SELECT DISTINCT ?nameartist ?paintingname ?
namemuseum ?museumcountry ?image
WHERE {
 ?artist rdf:type ex:Artist.
 ?artist ex:hasName ?nameartist.
 ?painting ex:paintedBy ?artist.
 ?painting ex:hasName ?paintingname.
 ?artist ex:birthCountry ?birthcountry.
   FILTER(?birthcountry='Italy')
 ?painting ex:situatedIn ?museum.
 ?museum rdf:type ex:Museum.
 ?museum ex:hasName ?namemuseum.
 ?museum ex:museumCountry ?museumcountry
   FILTER(?museumcountry!='Italy')
 ?painting <http://example.org/imageURL> ?image
}
```

Query 2

What are the other painting made by the painter who painted a specific painting?

Benois Madonna

Leonardo da Vinci

Virgin of the Rocks

Leonardo da Vinci

Adoration of the Magi

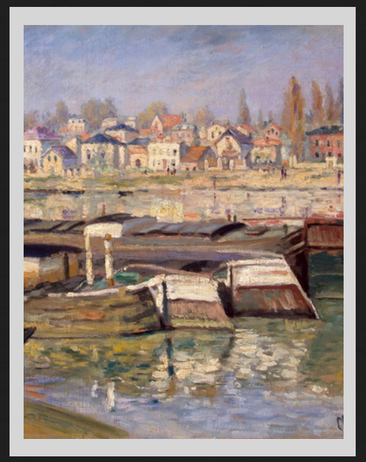
Leonardo da Vinci

```
SELECT DISTINCT ?nameartist ?paintingartistname ?image
WHERE {
 ?artist rdf:type ex:Artist.
 ?artist ex:hasName ?nameartist.
 ?painting ex:paintedBy ?artist.
 ?painting ex:hasName ?paintingname.
   FILTER(?paintingname='Mona Lisa')
 ?paintingartist rdf:type ex:Painting.
 ?paintingartist ex:hasName ?paintingartistname.
 ?paintingartist ex:paintedBy ?artist.
  ?paintingartist <http://example.org/imageURL> ?image
}
```

Query 3

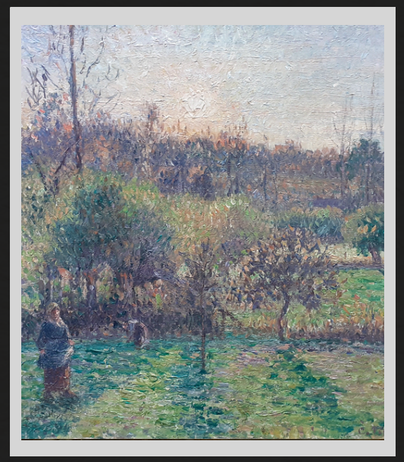
What paintings were painted in a specific artistic movement (impressionism)?

Landscape. Seine at Asnieres



Claude Monet

À Eragny, dans mon pré, soleil couchant



Camille Pissarro

Kids

Louise Catherine Breslau

```
SELECT DISTINCT ?movement ?paintingname ?nameartist ?image
WHERE {
 ?artist rdf:type ex:Artist.
 ?artist ex:hasName ?nameartist.
 ?painting ex:paintedBy ?artist.
 ?painting ex:hasName ?paintingname.
 ?painting ex:movement ?movement.
   FILTER(?movement='Impressionism')
  ?painting <http://example.org/imageURL> ?image
}
```

Challenges and Future Work

Challenges

- The constant rise of inconsistencies and missing data that were present in the data despite trying to avoid them in the queries and modelling phases.
- The difficulty of setting up the Talk to Your Graph agent due to the complicated procedure as well as Open AI issues.

Future Work

- The Knowledge graph could be enriched with descriptions and information to obtain a complete graph.
- The use of some visualisation tool such as a website in order to allow a smoother navigation through the knowledge graph for technical and non technical users.

Thank You!